

Predicting Pokémon Legendary Status: A Machine Learning Approach

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Problem Definition:

- Legendary Pokémon are special creatures designated by game creators without explicit criteria
- No clear formula exists for what makes a Pokémon legendary
- **The Goal:** Use machine learning to uncover patterns and relationships that might explain the legendary status designation

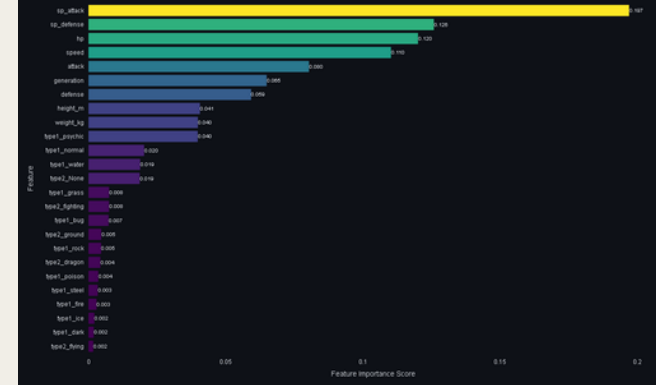
Dataset:

- The Complete Pokémon Dataset (Rounak Banik)
- 801 Pokémon entries across 8 generations
- **Features:**
 - Base stats (HP, Attack, Defense, Special Attack, Special Defense, Speed)
 - Physical characteristics (Height, Weight)
 - Type information (Primary and Secondary types)
 - Generation number
 - Capture rate
 - Experience points
 - Abilities
- **Target:** Binary Classification (Legendary vs Not)

Technical Approach:

- **Data Preprocessing**
 - Handled missing values
 - Normalized numerical features
 - Engineered new features (Total Stats, Type combinations)
 - **Model Selection**
 - Random Forests:
 - Handle numerical and categorical data
 - Capture non-linear relationships
 - Provide feature importance insights
 - Handle class imbalance effectively
 - **Class Imbalance Handling**
 - Original dataset: ~7% legendary Pokémon
 - Used SMOTE -> synthetic minority oversampling
 - Stratified sampling for training/test split
- ## Results:
- **Statistical Patterns**
 - Legendary Pokémon have higher base stats
 - Some features relate much more heavily to Legendary than others
 - Certain type combinations are more common among Legendary Pokémon

Top 25 Feature Importance



Model Performance

- 92% accuracy on test set
- High precision for legendary class
- Balanced recall across both classes

Insights

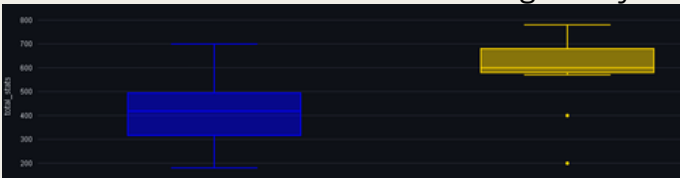
- Creators appear to use a combination of:
 - High base stats
 - Specific type combinations
 - Physical characteristics
- Special Attack and Defense have the Highest importance to Legendary

Further Work:

- Experiment with different classification models to compare performance
- Implement more advanced class imbalance techniques to improve minority class prediction
- Develop ensemble model methods for more robust predictions

Total Stats Distribution

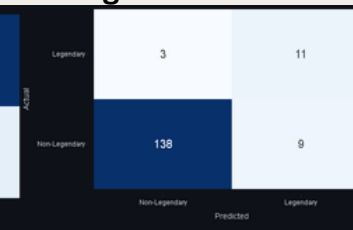
Common Legendary



Training Results



Testing Results



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